

# NAVADE HAQ

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## EDUCATION

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**Georgia Institute of Technology**, Atlanta, GA

Expected Graduation: December 2028

GPA: 4.0

Bachelor of Science in Electrical Engineering — Honors Program

Coursework: Digital Design, Circuit Analysis, Python

## RESEARCH EXPERIENCE

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**Compound Semiconductor Products VIP – Georgia Tech**

January 2026 – Present

Research team focused on GaN device design, simulation, fabrication, and characterization for implementation in analog circuits in partnership with Sandia National Laboratories

- Performing I-V characterization of Gallium Nitride High-Electron-Mobility Transistors (GaN HEMTs) utilizing a curve tracer.
- Matching HEMTs based on electrical behavior to evaluate wafer-level consistency.
- Simulating device models in QSPICE using measured curve-tracing data.

**Inan Lab – Georgia Tech**

September 2025 – Present

Multidisciplinary research group focused on physiological signal processing and the design of non-invasive health monitoring technology.

- Processed real-world PPG and ECG physiological signals using Python, applying Butterworth band-pass filters (0.5-40 Hz) to eliminate baseline wander and noise.
- Estimated heart rate in the frequency domain using Welch's method and peak detection.

## PROFESSIONAL EXPERIENCE

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**Yanfeng Automotive Interiors – Manufacturing Intern**, Romulus, MI    May 2025 – June 2025

- Troubleshoot defective Sierra EV and Silverado EV door latches using root cause analysis.
- Supported factory floor reorganization through interpretation of plant layout schematics.
- Worked on assembly line, developing an end-to-end understanding of manufacturing processes.

## SKILLS

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**Software:** SPICE, Cadence Xcelium, Cadence SimVision

**Hardware:** Curve Tracer, Oscilloscope

**Programming:** SystemVerilog, Python, MATLAB

## PROJECTS

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**64-bit SystemVerilog Calculator**

January 2026

- Designed 64-bit adder in SystemVerilog through instantiation of 32-bit adders.
- Developed 8-state finite state machine controller to sequence SRAM reads, adder operations, result buffering, and memory writes.
- Integrated controller, adder, buffer, and SRAM modules at chip top level.
- Verified design using Cadence Xcelium and SimVision, completing 128 64-bit additions in 773 cycles.